

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Built a multi-agent GenAI system via OpenAI Agents SDK and MCP tool-use, integrating ML models into production workflows handling millions of daily transactions at 99%+ uptime.
- Deployed a production RAG pipeline on AWS SageMaker for real-time decision support at sub-200ms p99 latency with RTO under 2 minutes.
- Engineered a pluggable distributed executor routing pipeline nodes across Docker and AWS Lambda, backed by an S3 artifact store eliminating redundant recomputation.
- Owned full-stack delivery of AI-powered platform features (React/TypeScript, Python/FastAPI); replaced polling with SSE for live ML metrics streaming, cutting redundant API calls 90% and enabling real-time model governance dashboards.
- Designed and optimized agentic AI workflows across multiple LLMs (GPT, Claude, Llama, Gemini), selecting and tuning models per use case for production quality and cost.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Cut p99 inference latency 420ms to 140ms across 50K+ daily requests by migrating synchronous model serving to async vLLM on GCP Cloud Run with autoscaling.
- Built PySpark data processing jobs for large-scale feature and dataset preparation feeding downstream model training and evaluation pipelines.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Built and operated 25+ production microservices in Java, C++, and SQL-backed distributed systems for 1M+ users, sustaining 99.9% uptime over 3 years on AWS (EC2, RDS, SQS).
- Cut nightly financial data aggregation blocking downstream settlement from 8 minutes to 90 seconds by redesigning SQL with indexed views and batch partitioning.
- Standardized failure-handling contracts across 25+ financial services, reducing production escalations 60% in one quarter via consistent retry, timeout, and propagation semantics.

PROJECTS

Distributed KV Store / VPC Control Plane Simulator

github.com/ganeshhgupta/distributed-kv-store

- Raft-consensus distributed KV store handling concurrent writes with linearizable reads and log compaction; companion VPC control plane simulator modeling subnet routing and security group enforcement.

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- LangGraph multi-agent system diagnosing production incidents via tool-use chains; routes alerts to specialized sub-agents and synthesizes root cause reports.

LangFetch

tinyurl.com/LangFetch

- LangGraph agent decomposes multi-table questions into dependency-ordered SQL subqueries with intermediate result passing; deployable as an MCP-compatible server for enterprise data queries.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*
MCKV Institute of Engineering | *B.Tech Computer Science*

Aug 2023 – May 2025
Jul 2016 – Jul 2020

SKILLS

Languages: Python, Java, C++, TypeScript, SQL

AI/ML & Agentic Systems: LLMs, RAG, MCP, OpenAI Agents SDK, LangGraph, PyTorch, vLLM, Hugging Face

ML Infrastructure: Model Deployment, Model Evaluation, Model Optimization, Data Processing, AWS SageMaker

Distributed Systems & Data: Distributed Systems, PostgreSQL, Redis, BigQuery, NoSQL

Cloud & DevOps: AWS (EC2, S3, SageMaker, Lambda), GCP (Cloud Run, Vertex AI), Docker, Kubernetes, CI/CD

CERTIFICATIONS

GCP Professional MLOps Engineer | AWS Certified Developer Associate